HOME / SERVICES / TEST AUTOMATION & VALIDATION

SERVICE 03 / EVIDENCE AND TEST

Tests worth *repeating.*

Connect instruments, data and analysis so a technical result can be reproduced, reviewed and used to choose the next test.

HIL AUTOMATION INSTRUMENTATION DATA PIPELINES

01 / PROBLEMS THIS ADDRESSES

When the result *cannot be repeated.*

Measurements become difficult to trust when procedures are informal, instrument settings are unclear or data analysis starts after the test instead of with it.

Typical starting points

> Scattered logs, manual test steps, missing operating-point context, instrument integration questions or a design decision that needs better evidence.

02 / TECHNICAL CAPABILITIES

Turn raw data into *usable evidence.*

Automation is shaped around the question the test needs to answer, not around a generic dashboard.

ACQUISITION

Instruments and channels

DAQ workflows, CAN and encoder logs, instrument control and structured test inputs where the system supports them.

ANALYSIS

Results with conditions

Plots, efficiency maps, error analysis, Allan deviation or stability measures when the data supports the claim.

03 / ENGAGEMENT WORKFLOW

Make the test *worth repeating.*

01

Define the question

State the decision and the measurable output.

02

Map the channels

Connect instruments, signals and operating points.

03

Run a baseline

Capture the current behavior before changing it.

04

Automate the repeat

Make the useful procedure consistent.

05

Report the boundary

State conditions, findings and the next test.

04 / REPRESENTATIVE EVIDENCE

Validation across *working systems.*

These public projects show hardware, firmware and measured evidence in different forms. They are not presented as completed client contracts.

CONTROL PLATFORM

ESP32 Robotics Control Platform

MATLAB/Simulink integration with IMU sensing, encoder feedback, motor control and self-balancing PID behavior.

VIEW PROJECT VIEW REPOSITORY ↗

POWER PROTOTYPE

5V / 1A SMPS

Waveform captures and bench validation notes used to review an offline flyback prototype.

VIEW PROJECT VIEW REPOSITORY ↗

UNIVERSITY RESEARCH

TI Piccolo PMSM dynamometer

Real-time testing, efficiency mapping, validation plots and dual-motor control integration.

VIEW PROJECT VIEW REPOSITORY ↗

05 / DELIVERABLES

Leave behind *usable evidence.*

- Automated test scripts or procedure definitions
- Instrument and data-acquisition configuration notes
- Structured logs, plots and analysis notebooks
- Validation reports with stated conditions
- Prioritized findings for the next test

> **Evidence follows access.**
>
> The method depends on the hardware, instrumentation, data quality and measurement access available.

START A TECHNICAL DISCUSSION

Make the result *reviewable.*

Describe the system, the current test method, what is hard to reproduce and which decision the evidence needs to support.

START A TECHNICAL DISCUSSION →

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